Duwon Yang

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Research Interests

Thick-panel origami · Reconfigurable and deployable structures · Kinematics of multi-DOF folding mechanisms · Multistability and buckling-induced actuation

Education

Pohang University of Science and Technology (POSTECH) *Pohang, Korea*
M.S. in Mechanical Engineering, Extreme Mechanics Lab (Advisor: Prof. Anna Lee) *Mar. 2025 – Feb. 2027 (expected)*

- GPA 4.10 / 4.30 · Thesis defense scheduled Nov. 2026
- Thesis: *Extended Offset Crease Method for Bi-Directional Folding of Thick-Panel Origami*

Handong Global University *Pohang, Korea*
B.S. in Mechanical & Control Engineering *Mar. 2019 – Feb. 2025*

- GPA 3.94 / 4.50 (major 4.14 / 4.50) · **Ranked 2nd out of 20 in the department**
- Mandatory military service, Republic of Korea, 2020 – 2021

Research Experience

Extreme Mechanics Lab, POSTECH *Pohang, Korea*
Graduate Research Assistant (Advisor: Prof. Anna Lee) *Mar. 2025 – Present*

- Developed an **extended offset crease method** admitting bidirectional folding at a thickened degree-6 vertex, eliminating the thickening and widening that accumulate across degree-4 vertex arrays
- Derived the kinematics of the multi-DOF mechanism through an **energy-based formulation** in MATLAB, resolving branch selection at the bifurcation of the fully unfolded state
- Demonstrated at least **182 reachable configurations** — 140 from module-shape combinations (70 shapes under C_4 symmetry × 2 folding directions) and 42 counted on diamond-pattern recessed pyramidal origami prototypes I fabricated and validated
- Designed and fabricated a **membrane-based Kresling pneumatic actuator** for a propellant-driven micro actuator (Lab-on-PCB), characterized under compression testing

Human Robotics Lab, Handong Global University *Pohang, Korea*
Undergraduate Researcher (Advisor: Prof. Jaehyo Kim) *2022 – 2024*

- Designed the experimental protocol, ran the measurements, and analyzed shoulder movement patterns under elastic resistance in relation to proprioceptive feedback (2023 – 2024); **co-first author** on the resulting *Bioengineering* paper

Project Experience

SeAH Wind Ltd., industry-funded project with Extreme Mechanics Lab, POSTECH *Pohang, Korea*
Research Participant *May 2025 – Feb. 2026*

- Modeled the deformation of a 105 m, 2,000-tonne offshore wind monopile on three turning-roll supports with an **Abaqus shell model** (quadratic shell elements, S355ML with plasticity, frictionless roller contact, 1.52g design load per DNV); the results fed into **revised support arrangements for the grow line**

Publications

1. **[In preparation] Yang, D.**, Kim, S., & Lee, A.* (2026). Diamond pattern-based bi-foldable thick panel origami for reconfigurable structures. *Thin-Walled Structures*.

2. **[Published]** Shim, G.[+], **Yang, D.[+]**, Cho, W., Kim, J., Ryu, H., Choi, W., & Kim, J.* (2024). Elastic resistance and shoulder movement patterns: an analysis of reaching tasks based on proprioception. *Bioengineering*, 11(1), 1. doi:10.3390/bioengineering11010001

3. **[Published]** Kwak, H.[+], Kim, J.[+], **Yang, D.**, Jun, G., Cho, Y., & Lee, A.* (2026). Development of mathematical and finite element models for predicting work-roll deformation in a 4-high hot rolling mill. *Transactions of Materials Processing*, 35(1). doi:10.5228/KSTP.2026.35.1.19

4. **[In preparation]** Lee, J.[+], Kim, S.[+], **Yang, D.**, Kim, M., & Lee, A.* Design and development of a propellant-driven micro pneumatic actuator using Lab-on-PCB technology.

[+] Equal contribution. * Corresponding author.

Conference Presentations

1. **Oral — Yang, D.** & Lee, A.* (2026). Diamond pattern-based bi-foldable thick panel origami for reconfigurable structures. *ASME International Mechanical Engineering Congress and Exposition (IMECE)*, Vancouver, Canada. (upcoming, Nov. 2026)

2. **Oral — Yang, D.**, Kim, S., & Lee, A.* (2026). Design methodology for bi-directionally foldable thick-panel origami. *KSME CAE and Applied Mechanics Division Spring Conference*, Korea.

3. **Poster — Yang, D.** & Lee, A.* (2025). Thickness-accommodating method for multi-configurable thick origami with bidirectional folding. *International Conference on Precision Engineering and Sustainable Manufacturing (PEESM)*, Chiang Mai, Thailand.

Honors & Awards

2025	**POSTECHIAN Fellowship**	*POSTECH*
2022	**Grand Prize**, Clutch the Energy Camp	*Ministry of SMEs and Startups, Korea*

Teaching Experience

Spring 2025	**Teaching Assistant**, Solid Mechanics	*POSTECH*
Spring 2024	**Teaching Assistant**, Linear Algebra	*Handong Global Univ.*
Fall 2023	**Teaching Assistant**, Solid Mechanics; Differential Equations	*Handong Global Univ.*

Outreach & Service

Origami Creativity Camp — Instructor *Pohang, Korea*
Taught folding geometry to elementary school students *Mar. – May 2026*

- Designed and ran nine weekly sessions for two age-grouped classes (12 students), translating my own research on folding mechanics into hands-on activities

Appropriate Technology Project (CRAIST90%) — Project Member *Abidjan, Côte d'Ivoire*
Charcoal briquettes from discarded coconut shells bound with cassava residue; *Mar. – Dec. 2019*
funded by a university grant for student-led international projects, in partnership with Food for the Hungry

Student Representative, Dept. of Mechanical Engineering, Handong Global University *2023*

Technical Skills

CAD & Graphics SolidWorks · AutoCAD · Illustrator · Photoshop

CAE Abaqus · Ansys

Programming MATLAB · C · Python · LaTeX

Experimental Laser cutting · molding · mechanical testing (Instron) · 3D printing